# VaRiA: Variable-Resolution Image Adaptive Robust Watermarking

Guanjie Wang, Zehua Ma, Han Fang, Chang Liu, Weiming Zhang, and Nenghai Yu

***Abstract*—Digital watermarking embeds imperceptible information into images to support copyright protection and source tracing. While recent deep learning-based watermarking methods achieve strong robustness under various distortions, they typically assume fixed input resolutions. However, images often appear in diverse sizes and aspect ratios, and existing extensions such as residual scaling or block-wise embedding offer only partial solutions, leading to degraded robustness, reduced visual quality. To address these challenges, we propose VaRiA (Variable-Resolution Image Adaptive robust watermarking), a Transformer-based framework that models pixel relationships consistently across resolutions, while a dual-stage training strategy first ensures robustness on fixed resolutions and then adapts to variable resolutions. Experiments demonstrate that VaRiA achieves nearly 100% watermark extraction accuracy under diverse distortions and resolutions, while maintaining high image fidelity (44 dB PSNR), confirming its robustness and practical value.**

***Index Terms*—Robust image watermarking, image processing, deep learning, intellectual property rights protection.**

## I. Introduction

DIGITAL watermarking technology achieves copyright protection and source tracing by embedding imperceptible authentication information into images. In recent years, the rapid advancement of image-sharing platforms and artificial intelligence generation techniques has significantly heightened interest in watermark research. Watermarking techniques not only enable the forensic analysis of unauthorized use but also help identify artificially generated images and trace their origins. Based on deep learning techniques, deep watermarking techniques [1], [2], [3], [4], [5] have achieved high watermark verification accuracy with slight modifications to cover images, even under various complex distortions. However, *this performance is typically achieved under the assumption that the input image resolution is fixed*. As shown in Table I, existing representative watermarking methods are usually trained and inferred with fixed-resolution square images, which commonly have a relatively low resolution ($< 512 \times 512$ pixels).

In practical applications, digital images often have varying resolutions. As shown in Table II, a survey of popular image-sharing platforms reveals that high-resolution images with unequal aspect ratios are the most prevalent. Additionally, AI-generated images [6], [7], [8] can have variable resolutions via outpainting. A notable gap exists between the high-resolution demands of practical scenarios and the resolutions currently supported by deep watermarking methods (see Table I).

TABLE I
Applicable Image Resolution of Representative Deep Watermarking Methods. $N^2$ Means $N \times N$ Pixels

| Method | HiDDeN [1] | MBRS [9] | CIN [4] | SSLWM [5] | MuST [3] | Stegastamp [2] |
|---|---|---|---|---|---|---|
| Resolution | $128^2$ | $128^2$ | $128^2$ | $224^2$ | $256^2$ | $400^2$ |

TABLE II
The Percentage (%) Distribution of Images Across Different Resolutions in Various Image-Sharing Platforms and Libraries

| Platform[1] / Resolution[2] (R) | Canva | 500px | Pixso | VCG.COM | Shutterstock | Pexels |
|---|---|---|---|---|---|---|
| $R \leq 256^2$ | 15.31 | 0.0 | 10.98 | 0.0 | 0.0 | 5.77 |
| $256^2 \leq R \leq 1080^2$ | 12.04 | 0.35 | 8.71 | 0.0 | 1.07 | 4.79 |
| $1080^2 \leq R$ | 72.65 | 99.65 | 80.31 | 100.0 | 98.93 | 89.53 |

Certainly, deep watermarking methods can embed watermarks at variable-resolution images through some image processing strategies, such as the residual scaling method or block-wise embedding. As shown in Fig. 1, existing watermarking methods can successfully embed watermarks by scaling the watermark residual or dividing the image into blocks. However, it should be noted that these strategies are suboptimal for variable-resolution robust watermarking, as most existing approaches fail to adapt to the scaling residual technique, resulting in compromised visual quality and robustness. Moreover, the block-wise embedding strategy incurs significant computational time overhead while requiring the entire watermarked block to be available at the extraction stage, rendering the watermarked image vulnerable to geometric distortions in the absence of additional synchronization mechanisms. Geometric distortions, such as cropping, are widespread in social media image-stealing. We believe the main reason is that convolutional neural network-based watermarking methods struggle to fully

[1]The website of each platform is www.canva.cn, 500px.com.cn, pixso.cn, www.vcg.com, www.shutterstock.com, www.pexels.com.

[2]For the access limitation of platforms for free users, we used sampling surveys within the permitted range to obtain the results in the table.

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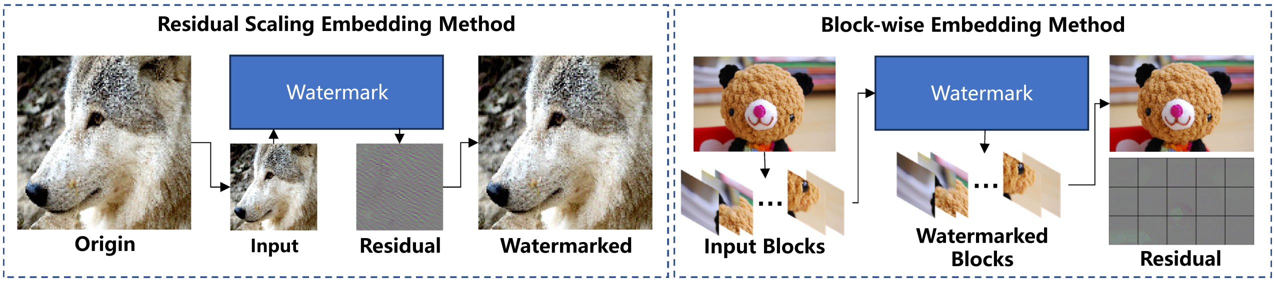



Fig. 1. The left figure shows a residual scaling method: the image is downsampled to $128 \times 128$, watermarked, then the residual is upscaled and added back to the original image. The right figure shows a block-wise method: the high-resolution image is split into patches, each embedded with the same watermark, and then reassembled. Black lines in the residuals highlight patch boundaries.

capture the relationship between image pixels, which is constant across variable resolutions, making it difficult to generate suitable watermark patterns for variable-resolution images. Furthermore, while some VAE [10]-based methods, like VINE [11], are capable of accommodating inputs within a certain resolution range, they exhibit suboptimal performance when processing inputs beyond the resolution scope of the VAE's pre-training data.

To address these issues, we propose VaRiA, an **Va**riable-**R**esolution **I**mage **A**daptive robust watermarking method. It is based on the Vision Transformer [12] architecture, which processes the target image by dividing it into patches. This design effectively captures the relationships between pixels across different patches, which is a feature that remains consistent across different resolutions. However, due to the lack of an effective watermark injection mechanism, we treat the watermark as a separate modality and incorporate it into the image features through an attention mechanism. To address the performance degradation caused by residual scaling, we propose a general two-stage training strategy. In the first stage, the model is trained on fixed-resolution images to achieve strong robustness and imperceptibility. In the second stage, the model is encouraged to capture the relative pixel relationships across images with varying resolutions, thereby mitigating the performance loss introduced by residual scaling. Experimental results indicate that VaRiA performs comparably to state-of-the-art (SOTA) methods and exhibits superior robustness in certain specialized application scenarios.

The main contributions of this paper are as follows:

- We propose VaRiA, a Transformer-based watermark framework for variable-resolution images. Leveraging attention mechanisms, VaRiA effectively captures pixel relationships and reconstructs watermarked images, ensuring strong performance across diverse scenarios.
- We introduce a dual-stage training strategy with an interpolation embedding and a patch-wise discriminator, enabling VaRiA to maintain robustness when processing images with variable resolutions.
- We treat the watermark message as a new modality, injecting it not only in the pixel domain but also through Cross-Attention mechanisms. This design reduces watermark robustness dependence on absolute pixel positions, ensuring strong performance against perturbations with pixel offsets while keeping good visual quality.
- Extensive experiments demonstrate the outstanding performance of VaRiA on both fixed-resolution and variable-resolution images and its practical usability in real-world scenarios.

## II. Related Works

In recent years, while robust watermarking techniques have focused on resisting a broader range of distortions, a lack of discussion of variable-resolution images has led to the limitation of applications of watermarking techniques in the real world. In this section, we first review studies focused on robustness against common digital distortions and unsupported variable-resolution images. Subsequently, we introduce research efforts aimed at addressing real-world channel distortions supporting variable-resolution images, such as those caused by screen-shooting attacks and social media dissemination.

### A. Watermarking Unsupporting Variable-Resolution Images

In recent years, with the development of deep learning, a series of works that aim to embed watermarks in images have been developed. HiDDeN [1], proposed by *Zhu* et al., is the first end-to-end training framework. HiDDeN jointly trains an encoder, which is used to embed the watermark into the cover image, a noise layer, which simulates common digital distortions, and a decoder, which extracts the watermark from the distorted image. Based on the encoder-decoder framework, MBRS [9] utilizes the Squeeze-and-Excitation blocks [13] and proposes a message processor to expand the message more effectively, aiming to enhance robustness against JPEG compression. *Fang* et al. [14] simplified the noise layer by retaining only key distortions such as perspective transformation, illumination variation, and moiré patterns. Building on this, they proposed PIMoG, which achieved competitive performance. *Ma* et al. [4] proposed CIN, combined reversible and non-reversible mechanisms with a non-reversible attention module to address asymmetric extraction under noise attacks. Different from mainstream encoder-decoder architectures, SSLWM [5] embeds messages in the image features extracted by ResNet-50 [15] and extracts messages with a group of learnable secret keys. The robustness of SSLWM comes from data augmentation. MuST [3] extends the traditional encoder-decoder architecture by incorporating a watermark localization network, which identifies all watermarked regions within the composite image. It then applies a minimum bounding rectangle and connected component analysis to separate and verify each region. In contrast, WAM [16] eliminates the need for an explicit localization module by leveraging the powerful capabilities of SAM [17], enabling simultaneous localization and extraction of watermarks. *Hu* et al. [18] proposed MaskMark, which, similar to MuST, consists of separate encoder, decoder, and locator modules. Unlike MuST, however, MaskMark fuses the image mask with the watermark features to produce the final watermark output. This design enables effective control over the watermarked regions within the image.

Although these watermarking methods exhibit excellent imperceptibility and robustness, they often fail to support variable-resolution images with practically viable robustness.

### *B. Watermarking Supporting Variable-Resolution Images*

Traditional watermarking algorithms are typically based on image transformations, such as DCT and DWT [19], or block-wise embedding, such as LGDR [20], which inherently allows them to handle inputs of variable resolutions. LGDR embeds symmetric watermarks in the spatial domain to obtain robustness against various geometric distortions and image processing operations. *Fang* et al. [21] proposed a novel screen-shooting resilient watermarking method that utilizes an intensity-based SIFT(Scale Invariant Feature Transform) algorithm to identify robust regions for embedding the well-designed watermark. Although traditional watermarking algorithms can achieve strong performance in certain specific scenarios, the hand-crafted features they rely on generally lack the generalization capability of features learned through deep learning methods, making it difficult for them to outperform deep learning-based approaches in overall performance.

Some watermarking algorithms support inputs with variable resolutions because they embed the watermark within specific local regions of the image. For example, LIM [22] hides information in a sub-image rather than the entire image and includes a localization module to correct the shooting distortions in the end-to-end framework. DIPW [23] is designed to enhance the robustness of watermarking methods under deliberate plagiarism. Similar to LIM, DIPW also hides copyright evidence in a patch determined by SIFT. However, these methods are inherently vulnerable to common distortions such as cropping. VideoSeal [24] introduces temporal watermark propagation, a technique to convert any image watermarking model to an efficient video watermarking model without the need to watermark every high-resolution frame.

Some watermarking methods also support inputs with variable resolutions by specifically adapting their designs, leveraging pixel-domain residual-based watermark embedding strategies. StegaStamp [2] decomposes the physical capturing process into a series of image processing operations to further enhance robustness to distortions resulting from real-world printing and photography. TrustMark [25] is the only deep learning-based watermarking solution currently designed to specifically address and optimize performance for different input resolutions. Although these methods can support inputs with variable resolutions, they lack optimization strategies specifically designed for handling images of varying resolutions. Moreover, they do not account for the fact that images with different resolutions often correspond to different application scenarios.

## III. Methods

In this section, we illustrate the proposed VaRiA architecture and the two-stage fine-tuning training strategy.

### *A. Architecture*

*a) Overall:* As illustrated in Fig. 2, VaRiA is an end-to-end asymmetric encoder-decoder architecture that incorporates a noise module between the encoder and decoder. This module introduces perturbations, such as scaling and Gaussian noise, serving as a data augmentation technique for the decoder. This enhances the decoder's robustness in extracting watermark messages across various disturbance scenarios. Inspired by GAN [26], a discriminator is trained concurrently to differentiate between watermarked and non-watermarked images. The encoder must generate sufficiently realistic watermarked images to deceive the discriminator, thereby optimizing the visual quality of the watermarked images produced by the encoder.

*b) Encoder:* As the first stage of the pipeline, the Encoder embeds the watermark message into the cover image to generate the encoded image. The design of the VaRiA encoder differs from all previous digital watermarking encoders. First, the encoder transforms the input image into $16 \times 16$ patches using convolutional layers, followed by a rearrangement operation. The watermark message is projected through a linear layer and concatenated with the image embeddings, which are then fed into an MAB layer. The resulting message attention is passed through another linear layer and concatenated with the image attention before being input to the next MAB layer. Finally, the output image feature map is concatenated with the original image and passed through a convolutional layer to produce the watermarked image.

To better capture the relationships between pixels at different resolutions and effectively utilize this information for watermark embedding, we propose the **Message Attention Block** (MAB), inspired by the cross-modality attention mechanism from multimodal models. MAB enables the model to adjust attention weights, which enhances its ability to robustly embed watermark information by leveraging spatial relationships. This mechanism ensures that the embedded watermark information is distributed more effectively across varying resolution levels, improving both visual quality and robustness. The formulation is as follows:

$$e = \mathbf{C}(e_m, e_I) \tag{1}$$

$$\hat{e_m}, \hat{e_I} = Split(LN(e)) \tag{2}$$

$$\hat{e_m^q}, \hat{e_m^k}, \hat{e_m^v} = W_m^q(\hat{e_m}), W_m^k(\hat{e_m}), W_m^v(\hat{e_m}) \tag{3}$$

$$\hat{e_I^q}, \hat{e_I^k}, \hat{e_I^v} = W_I^q(\hat{e_I}), W_I^k(\hat{e_I}), W_I^v(\hat{e_I}) \tag{4}$$

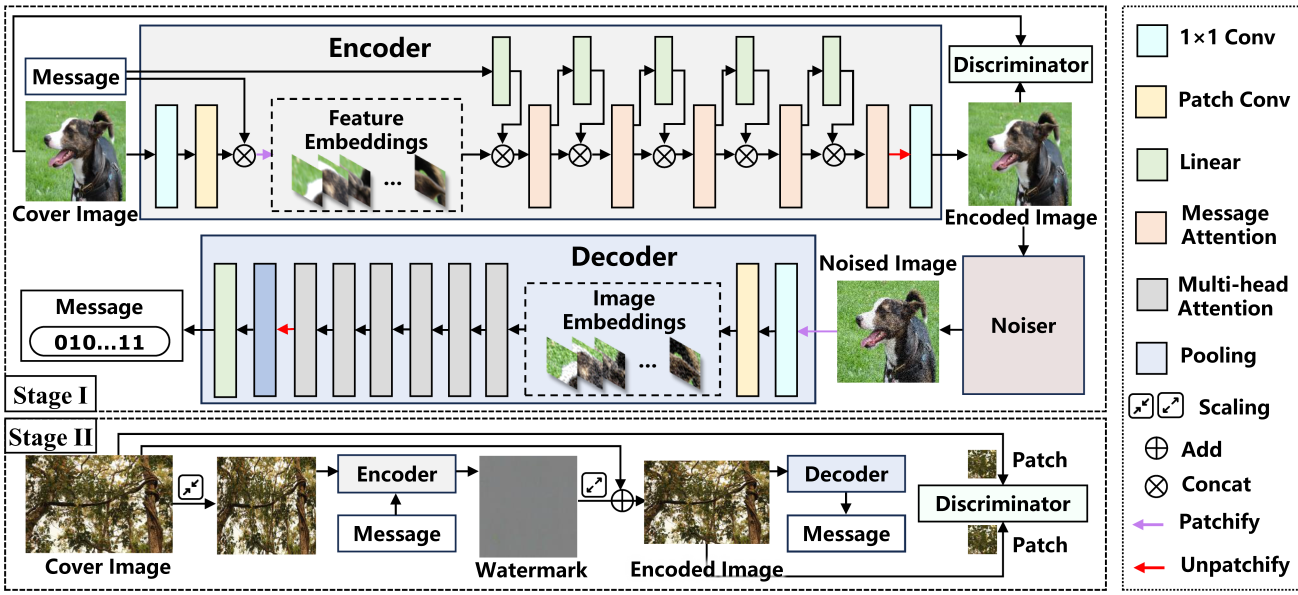



Fig. 2. The framework of the proposed VaRiA. The encoder embeds the watermark messages into cover images to generate the encoded images. An adversary discriminator is used to improve the visual quality of encoded images. Then, the decoder extracts the watermark message from the encoded image, which is enhanced by the noise module. VaRiA adopts a two-stage training strategy. It is trained on a substantial dataset of fixed-resolution images and then fine-tuned on variable-resolution images with an interpolation embedding technique and a patch-wise discriminator.

$$e^q, e^k, e^v = \mathbf{C}(\hat{e}_I^q, \hat{e}_m^q), \mathbf{C}(\hat{e}_I^k, \hat{e}_m^k), \mathbf{C}\left(\hat{e}_I^v, \hat{e}_m^v\right) \tag{5}$$

$$\hat{e} = Attention\left(e^q, e^k, e^v\right) + e \tag{6}$$

where $\mathbf{C}$ represents the tensor concatenation operation, $e$ denotes the message and image embeddings and $W$ signifies the linear mapping operation. Intuitively, the MAB facilitates cross-modal interaction step-by-step. It first concatenates the message and image embeddings (1) and stabilizes their distributions via Layer Normalization before splitting them back (2). Subsequently, both normalized features are independently projected into their respective Query, Key, and Value subspaces to extract intrinsic structural and visual patterns (3), (4). By concatenating these modality-specific representations (5), the model is guided to dynamically seek spatial alignments between the hidden message and the image content. Finally, a multi-head attention mechanism fuses these features, employing a residual connection to preserve the original visual context and prevent image degradation (6).

To constrain the generation quality of the encoder, we draw inspiration from GANs by employing a trainable discriminator for adversarial training of the encoder, aiming to optimize the visual quality. The loss function required for the encoder can be expressed as follows:

$$L_{reconstruct} = \frac{1}{n}\|\mathbf{I_{wm}} - \mathbf{I}\|_2^2 \tag{7}$$

$$L_{vgg} = \frac{1}{n}\|VGG_{16}(\mathbf{I_{wm}}) - VGG_{16}(\mathbf{I})\|_2^2 \tag{8}$$

$$L_{LPIPS} = LPIPS\left(\mathbf{I_{wm}}, \mathbf{I}\right) \tag{9}$$

$$L_{adversial} = \log\left(1 - DIS\left(\mathbf{I_{wm}}\right)\right) \tag{10}$$

where $n$ denotes the total number of pixels in the entire image, $L_{reconstruct}$ represents the reconstruction loss, ensuring that the watermarked image is close to the original image, and $L_{adversial}$ is the adversarial loss from the discriminator, which encourages the encoder to generate realistic watermarked images. $L_{vgg}$ and $L_{LPIPS}$ are also used to constrain the pixel domain's perceptual loss. $VGG_{16}(\cdot)$ extracts the feature maps from the relu2_2 layer (the second convolutional layer of the second block) of a VGG [27] network pre-trained on ImageNet. $LPIPS(\cdot)$ follows *Zhang* et al.'s work [28]. The final encoder loss $L_{Enc}$ is the weighted sum of these loss functions, which is formulated as follows:

$$\begin{aligned} L_{enc} = &\lambda_{reconstruct} L_{reconstruct} + \lambda_{vgg} L_{vgg} \\ &+ \lambda_{LPIPS} L_{LPIPS} + \lambda_{adversial} L_{adversial} \end{aligned} \tag{11}$$

*c) Discriminator:* To ensure high visual fidelity, the Discriminator adversarially evaluates the encoded image generated by the Encoder. We design a simple five-layer network based on ResNet [15] to serve as the discriminator. We introduce the adversarial loss $L_{adversial}$ to further enhance the visual quality of $I_{wm}$, which makes the encoder aim at generating an indistinguishable $I_{wm}$ by the additional adversarial discriminator.

Meanwhile, the discriminator $DIS$ aims to minimize the loss $L_{DIS}$:

$$\mathcal{L}_{DIS} = \log(DIS(\mathbf{I_{wm}})) + \log(1 - DIS(\mathbf{I})). \tag{12}$$

For the adversarial training, we adopt a straightforward 1:1 alternating update schedule. In each training step, the Discriminator and the Encoder-Decoder pipeline are updated sequentially.

*d) Noiser module:* To simulate real-world distortions and build robustness, the encoded image is then passed through the Noiser module. We introduce an adversarial perturbation Noiser module that applies disturbances to the input watermarked images. This setup enhances the robustness of the decoder during joint training, enabling it to effectively handle various distortions commonly encountered in images of different resolutions. Moreover, to ensure joint training of the encoder and decoder, all operations within this module are differentiable and support batch processing. The noised image can be formulated as follows:

$$I_{noised} = Noiser(I, choice). \tag{13}$$

$choice$ denotes the type of distortion applied to the image during the current training iteration.

*e) Decoder:* Finally, the Decoder receives the distorted image from the Noiser to extract the hidden watermark message. For the decoder, we did not adopt the same design as the encoder. Instead, we introduced several modifications inspired by the Vision Transformer (ViT [12]) architecture to better suit the requirements of the watermark extraction task. Specifically, the image is first passed through a $1 \times 1$ convolution layer and then partitioned into $16 \times 16$ patches. These patches are processed by a multi-head attention module, after which the features are reconstructed into a feature map and passed through a pooling layer to extract the watermark. Before being fed into the extractor, the image typically passes through the Noiser module to apply perturbations to encourage the extractor to learn robustness. Besides, to ensure that the decoder does not overfit the perturbations, we also extract messages from the clean watermarked images. The loss for the decoder is formulated as follows:

$$L_{dec} = \frac{1}{n}\|\mathbf{m_{noised}} - \mathbf{m}\|_2^2 + \frac{1}{n}\|\mathbf{m_{clean}} - \mathbf{m}\|_2^2 \tag{14}$$

where $n$ denotes the total number of bits in the entire message.

### B. Training Strategy

We employed a two-stage training strategy to ensure that the algorithm performs well on standard low-resolution images while also maintaining good visual imperceptibility and robustness when processing variable-resolution images.

*a) Stage I:* In the first stage, we follow the classic end-to-end training approach for robust watermarking algorithms, where the encoder and decoder are jointly trained. After the input image passes through the encoder to embed the watermark message, it is processed by the adversarial noise layer and then fed into the decoder to extract the correct watermark. This joint training couples the encoder and decoder, allowing the encoder's parameters to be updated with the awareness of the decoder's computations, and vice versa. This mutual influence helps the model better achieve the goal of embedding a robust watermark with minimal perceptual changes to the original image.

Finally, we can formulate the loss function for the training stage I as follows:

$$Loss_I = L_{enc} + \lambda_{dec} L_{dec} \tag{15}$$

*b) Stage II:* The goal of the second stage of training is to fine-tune the model's parameters so that it maintains strong performance in low-resolution scenarios while adapting to variable-resolution images. To achieve this, we introduce an interpolation embedding technique and a patch-wise discriminator in the training process based on Stage I. As shown in Fig. 2, we first scale the image to a fixed resolution, which is the same as Stage I, to serve as input, then pass it through the encoder to obtain the watermark. The watermark is then resized to the original image resolution and added to the image. If we continue to constrain visual loss across the entire image as in Stage I, this would require performing the calculation on the resized image; otherwise, the computational cost becomes prohibitive. However, resizing may remove image detail, potentially causing the final watermark to lack focus on finer details.

Inspired by PatchGAN [29], we randomly crop patches of the same location from both the original and watermarked images and use these patches to constrain the visual loss. After sufficient training steps, the model can focus on details across the entire image while minimizing visual loss. Therefore, we can formulate the loss function for the training stage II as follows:

$$Loss_{II} = \sum_{i=1}^{n} L_{enc}(I_{enc}^i, I^i)/n + \lambda_{dec} L_{dec} \tag{16}$$

where $n$ represents the number of image patches for computing loss. In order to balance computational efficiency and loss function performance, we set $n = 4$.

## IV. Experiments

### A. Experiment Settings

*1) Datasets:* In the first stage of training, we use the MS COCO2017 [30] dataset, which includes approximately 10 k images. The second stage involves fine-tuning the model on the DIV2K [31] dataset, including a total of 800 2 k resolution images in the training set. To evaluate the model's performance after the first stage, we use 1000 images from the MS COCO2017 validation set, while the results of the second stage are validated on 400 images from the DIV2K validation set and the Flickr2K [32] dataset. It's worth noting that we scaled images into $224 \times 224$ for stage I and kept the original resolution for stage II.

*2) Implementation Details:* The whole framework is implemented by PyTorch [33] and executed on a NVIDIA RTX A6000 GPU. Stage I takes 7 days with one GPU. Stage II takes 1 d with one GPU. We utilize AdamW [34] as the optimizer. The watermark messages are randomly generated sequences of 64 bits. Input image size is fixed to $224 \times 224$, and the patch size for the transformer block is $16 \times 16$. The batch size in the training stage I is set to 128, and the VaRiA models are trained for 500 epochs with an initial learning rate $= 0.0001$. The batch size in the training stage II is set to 16, and the VaRiA models are trained for 300 epochs with an initial learning rate $= 0.0001$. The parameter settings used in the loss function are, for training stage I $\lambda_{reconstruct} = 9.0, \lambda_{vgg} = 1.0, \lambda_{LPIPS} = 5e-2, \lambda_{adversial} = 1e-2, \lambda_{dec} = 16.0$ and

TABLE III
THE AVERAGE $\overline{ACC}(\%)$ RESULTS TESTED ON THE FIXED RESOLUTION UNDER THE CLASSIC DISTORTION SETTINGS. BOLD RESULTS INDICATE THE BEST OUTCOME AMONG THE COMPARISONS, WHILE UNDERLINED RESULTS SIGNIFY THE SECOND-BEST OUTCOME.

| Type | Identity | RealCrop | Crop | Cropout | Dropout | Resize | JPEG | GB | GN | Brightness | Contrast | Saturation | Grayscale |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Settings | -<br>- | 30%<br>70% | 30%<br>50% | 9%<br>30% | 30%<br>50% | 0.4<br>1.2 | $Q=40$<br>$Q=80$ | $\sigma=0.5$<br>$\sigma=2$ | $s=0.25$<br>$s=1.0$ | 0.5<br>2 | 0.5<br>2 | 0.5<br>2 | -<br>- |
| LGDR | 97.88 | 54.22 | 58.55 | 97.88 | 97.88 | 62.65 | 95.86 | 51.47 | 91.48 | 96.61 | 96.41 | 97.63 | 70.33 |
| HiDDeN | 99.99 | 53.87 | 89.64 | 98.15 | 94.59 | 91.26 | 80.35 | 99.51 | 99.19 | 98.15 | 98.59 | 99.51 | 79.26 |
| SSLWM | 99.99 | 57.81 | 97.74 | 99.32 | 97.65 | 94.41 | 96.74 | 99.13 | 97.41 | 95.86 | **99.84** | <u>99.84</u> | <u>93.31</u> |
| CIN | 99.99 | 56.39 | 98.32 | **99.99** | **99.99** | 89.64 | **99.72** | <u>99.94</u> | **99.99** | <u>99.13</u> | 99.58 | **99.92** | 90.71 |
| MuST | 99.16 | <u>84.08</u> | **99.16** | 99.51 | 98.93 | <u>97.34</u> | 96.29 | 98.13 | 99.16 | 98.79 | 98.38 | 98.47 | 81.59 |
| VaRiA | **99.99** | **98.72** | <u>99.27</u> | <u>99.75</u> | **99.99** | **99.74** | <u>99.65</u> | **99.99** | <u>99.92</u> | **99.92** | <u>99.73</u> | 99.71 | **98.44** |

TABLE IV
THE TABLE SHOWS THE CORRESPONDING WATERMARK CAPACITY FOR EACH METHOD. $*$ REPRESENTS THAT WE UTILIZED A ROBUST SPREAD SPECTRUM CODING METHOD.

| Methods | LGDR | DCTDWTSVD | HiDDeN | MuST | SSLWM | CIN | TrustMark | TrustMark* | VideoSeal-0.0 | VideoSeal-0.0* | VideoSeal-1.0 | VideoSeal-1.0* | VaRiA |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Length(bits) | 30 | 64 | 30 | 30 | 30 | 30 | 100 | 50 | 96 | 48 | 256 | 64 | 64 |

TABLE V
THE AVERAGE $\overline{ACC}(\%)$ RESULTS TESTED ON THE VARIABLE RESOLUTION UNDER DISTORTION SETTINGS. I INDICATES THAT THE MODEL WAS TRAINED AT A FIXED RESOLUTION. II INDICATES THAT THE MODEL WAS FINE-TUNED ON IMAGES OF VARIABLE RESOLUTIONS. BOLD RESULTS INDICATE THE BEST OUTCOME AMONG THE COMPARISONS, WHILE UNDERLINED RESULTS SIGNIFY THE SECOND-BEST OUTCOME.

| Type | Crop | RealCrop | Resize | JPEG^ | GB | GN | Brightness | Contrast | Saturation | Grayscale | Logo | Padding |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Settings | 30%<br>70% | 30%<br>50% | 0.4<br>1.2 | $Q=40$<br>$Q=80$ | $\sigma=0.2$<br>$\sigma=2$ | $s=0.25$<br>$s=1.0$ | 0.5<br>2 | 0.5<br>2 | 0.5<br>2 | -<br>- | $num=10$<br>$num=20$ | 720:1080<br>1440:2560 |
| DCTDWTSVD | 59.19 | 48.77 | 80.86 | 83.5 | 82.08 | 81.89 | 78.13 | 81.95 | 85.94 | 78.94 | 55.64 | 51.72 |
| HiDDeN-I | 51.31 | 53.87 | 54.27 | 53.88 | 53.16 | 57.11 | 55.97 | 55.97 | 58.64 | 53.08 | 53.25 | 52.61 |
| MuST-I | 56.17 | 53.51 | 52.49 | 56.17 | 57.63 | 65.57 | 62.84 | 55.31 | 56.17 | 51.82 | 58.66 | 64.29 |
| TrustMark-Q | 98.01 | 51.24 | 98.33 | 94.26 | 99.02 | 98.50 | 99.05 | 99.44 | 99.17 | 99.31 | 90.72 | 81.02 |
| TrustMark-Q* | <u>98.96</u> | 54.00 | 98.71 | <u>96.60</u> | 99.14 | <u>99.57</u> | <u>99.36</u> | <u>99.44</u> | **99.68** | **99.68** | <u>85.29</u> | 84.15 |
| TrustMark-P | 81.73 | 51.90 | 94.44 | 91.47 | 89.60 | 86.84 | 89.01 | 92.39 | 93.94 | 84.98 | 71.58 | 74.32 |
| TrustMark-P* | 84.28 | 50.46 | 96.85 | 95.08 | 89.94 | 90.10 | 91.42 | 93.56 | 97.28 | 85.34 | 70.46 | 75.03 |
| VideoSeal-0.0 | 70.67 | 83.78 | 95.71 | 66.35 | 96.11 | 96.43 | 65.63 | 91.77 | 95.97 | 50.40 | 60.42 | 58.71 |
| VideoSeal-0.0* | 72.39 | 84.06 | 96.81 | 68.73 | 97.92 | 96.86 | 67.56 | 93.97 | 96.42 | 51.15 | 63.08 | 57.80 |
| VideoSeal-1.0 | 53.78 | 86.10 | 99.51 | 76.53 | 99.63 | 76.42 | 68.91 | 97.74 | 99.16 | 98.63 | 82.67 | 85.06 |
| VideoSeal-1.0* | 58.96 | <u>88.24</u> | **99.92** | 80.09 | <u>99.93</u> | 79.15 | 71.30 | 98.61 | <u>99.66</u> | <u>99.18</u> | 83.91 | <u>87.25</u> |
| HiDDeN-II | 84.07 | 50.13 | 84.37 | 78.23 | 88.62 | 86.13 | 87.77 | 84.76 | 89.21 | 51.29 | 75.87 | 69.03 |
| MuST-II | 90.15 | 80.62 | 87.61 | 85.91 | 92.31 | 90.87 | 94.02 | 92.74 | 89.13 | 53.08 | 82.06 | 79.14 |
| VaRiA-I | 77.62 | 72.72 | 75.54 | 74.89 | 79.82 | 78.96 | 83.34 | 84.87 | 82.65 | 81.35 | 74.23 | 65.61 |
| VaRiA-II | **99.65** | **98.44** | <u>99.17</u> | **99.51** | **99.95** | **99.99** | **99.99** | **99.65** | 99.65 | 98.53 | **99.73** | **97.81** |

for training stage II $\lambda_{reconstruct} = 5.0, \lambda_{vgg} = 1.0, \lambda_{LPIPS} = 5e-2, \lambda_{adversial} = 1e-2, \lambda_{dec} = 12.0$. For noise in training, the parameter settings are as follows:

- *RealCrop:* $[30\%, 40\%]$
- *Brightness adjustment:* [0.5,2]
- *Contrast adjustment:* $[0.5, 2]$
- *JPEG compression adjustment:* [40,90]
- *Gaussian noise adjustment:* $\sigma \in [0.25, 0.75]$
- Grayscale

All of these noises are used with the same probability in each training step. For noise used in testing, the *Settings* in Table III and V describe the upper and lower limits of the noise range. In each test step, the noise intensity is sampled from the corresponding range according to a uniform distribution for testing, and the average performance is finally obtained.

*3) Baselines and Evaluation Protocols:* For deep-learning-based baselines, we choose HiDDeN [1] (the first deep watermarking method), SSLWM [5], CIN [4], and MuST [3], all of which are representative methods with fixed input and output resolutions. Moreover, TrustMark [25] and VideoSeal [24] is included for comparison with VaRiA training after Stage II due to their capability of handling variable resolutions. For classic methods that utilize artificial features, we choose LGDR [20] and DCTDWTSVD [35] for their effectiveness under various distortions.

Our testing framework is explicitly divided into two scenarios to comprehensively evaluate the baselines alongside our models (VaRiA for Stage I, and VaRiA-II for Stage II):

- *Fixed-Resolution Evaluation:* To assess the peak robustness of each watermarking method at its optimal supported resolution, we sample 1,000 images from the MS COCO2017 validation set.
- *Variable-Resolution Evaluation:* To rigorously test robustness and scaling invariance under arbitrary resolutions, we employ a mixed high-resolution dataset comprising 400 images from the DIV2K validation set and the Flickr2K dataset [32]. Notably, for baseline methods lacking native architectural support for variable-resolution inputs, we apply the residual scaling strategy by default to facilitate a fair and practical comparison in this setting.

**Algorithm 1:** Bitwise Payload Expansion Coding.

**Require:** input the capacity of watermark $C$
 $Real\_Length = C/2$
 $Watermark = ''''$
 **for** $i \leftarrow 1$ to $Real\_Length$ **do**
  $r = randomchoice(0, 1)$
  **if** $r = 0$ **then**
   $Watermark\ concate\ ''01''$
  **else**
   $Watermark\ concate\ ''10''$
  **end if**
 **end for**
 $Return\ Tensor(Watermark)$

**Algorithm 2:** Bitwise Payload Expansion Decoding.

**Require:** input watermark tensor $W$
 $W\_Divided = DivideByGroup(W,\ group\_size = 2)$
 $Watermark = ''''$
 **for** $group$ from $W\_Divided$ **do**
  **if** $group[0] < group[1]$ **then**
   $Watermark\ concate\ ''0''$
  **else**
   $Watermark\ concate\ ''1''$
  **end if**
 **end for**
 $Return\ Watermark$

When configuring watermark capacity, we note that the pretrained models released by TrustMark and VideoSeal have a fixed embedding capacity that is difficult to adjust. To ensure a fair comparison, we apply a **bitwise payload expansion** strategy to reduce their effective payload while enhancing robustness. This strategy is inspired by the redundancy principle of spread spectrum communications [36], [37], [38], yet it operates strictly at the data layer rather than distributing signal energy across the frequency spectrum. A detailed description of this coding algorithm are shown in Algorithm 1, 2. During encoding, each binary bit is mapped to a two-element sequence: a 0 b is encoded as the sequence [0, 1], and a 1 b as [1, 0]. During decoding, we utilize the message tensors output by models to compare the magnitudes of consecutive elements. More precisely, when the first magnitude in a consecutive pair is lower than the second, the corresponding bit is decoded as 0; otherwise, it is decoded as 1. Results shown in Table V demonstrate that the introduction of spread spectrum coding does not lead to a decrease in the robustness of watermarking. Finally, we conduct comparative experiments under settings shown in Table IV where our proposed method achieves better visual quality and higher capacity than the baseline.

*4) Metrics:* The peak signal-to-noise ratio (PSNR [39]) and structural similarity (SSIM [40]) are used to evaluate the visual quality of the encoded image materials. The average extraction accuracy ($\overline{ACC}$), i.e., the correct percentage of the extracted watermark message, is adopted to evaluate the robustness. The formulas used to calculate the above metrics are shown as follows:

$$\overline{(ACC)}(\%) = (\frac{1}{L} \times \sum_{i=1}^{L} 1 - |M_{dec}[i] - M[i]|) \times 100\% \tag{17}$$

$$PSNR(I_{wm}, I) = 20 \times \log_{10} \frac{MAX(I_{wm}, I) - 1}{\sqrt{MSE(I_{wm}, I)}} \tag{18}$$

$$SSIM(I_{enc}, I) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{19}$$

where $MAX(\cdot)$ is the maximum pixel value of images, and Symbol $\sigma$, $\mu_x$ and $\sigma_{xy}$ represent the average, variances and covariance of images, respectively. $C_1$ and $C_2$ are two constants for preventing unstable results.

## B. Experiment Results

*1) Fixed Resolution Robustness:* We compare the visual quality and the extraction accuracy after distortions at the optimal fixed image resolution required for each compared method. The visualizations of implemented distortions are shown in Fig. 3. VaRiA has the highest PSNR of 45.98 dB and SSIM of 0.9957 among all the compared methods; see Table VI. In Table III, VaRiA demonstrates results derived from models trained with a Noise module that combines all distortions shown in Fig. 3. Examples of watermarked images and watermark patterns are shown in Fig. 7. Notably, VaRiA achieves excellent robustness under *RealCrop* distortions, which are challenging for many watermarking methods. This robustness likely stems from the bidirectional attention mechanism, which reduces dependency on the absolute pixel positions by utilizing relative pixel positioning for watermark extraction. In contrast, other methods rely on convolutional networks (e.g., MuST, CIN) or manually crafted robust features (e.g., LGDR), where the convolutional kernels are more sensitive to absolute pixel positions, leading to subpar performance under such transformations. As can be seen, our proposed VaRiA not only ensures high visual quality but also demonstrates strong robustness against complex and diverse distortions.

*2) Variable Resolution Robustness:* Table VI presents the visual quality of VaRiA and baseline methods in variable resolution scenarios after being fine-tuned in stage II. II means we fine-tune models utilizing our proposed training stage II. The setting of *Padding* represents the noised image's height and width ratio. *JPEG*$^{\hat{}}$ represents the operation in which the image is downsized to the resolution of $256 \times 256$ first and then compressed and recovered to the original shape to cut down the computation cost. Table V presents the robustness of these methods in variable resolution scenarios before and after stage II. Since TrustMark and VideoSeal are already adapted to variable resolution scenarios during both training and testing, no additional fine-tuning was applied. Considering some common application scenarios for variable-resolution images, we introduced two types of perturbations: Logo and Padding, which are common in the application scenarios of variable-resolution

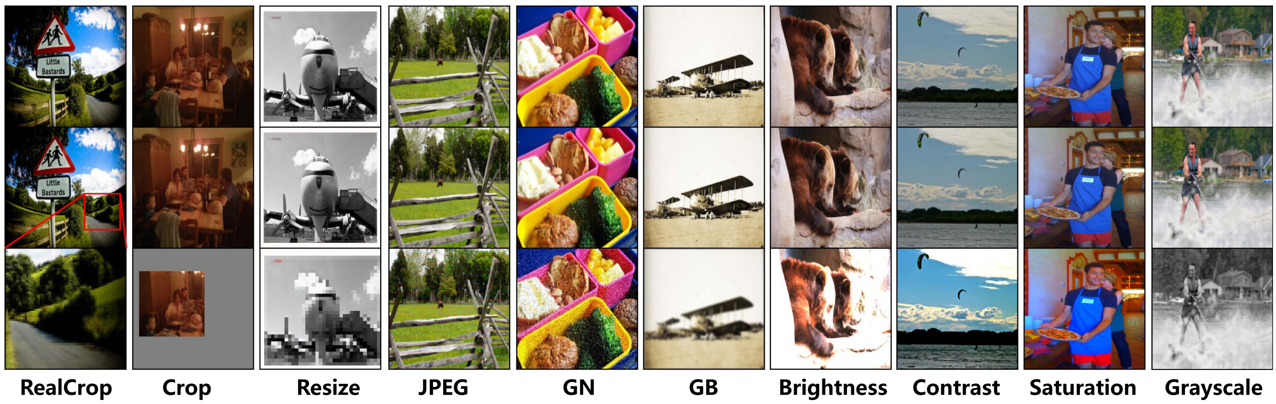



Fig. 3. Visual comparisons of experimental results under different classic noises. Top row: input image. Middle row: watermarked encoded image. Bottom row: noised image. GB: Gaussian Blur, GN: Gaussian Noise. RealCrop simulates partial cropping followed by resizing to the original shape.

TABLE VI
THE AVERAGE PSNR (dB) AND SSIM OF EACH METHOD TESTED ON THE FIXED RESOLUTION (F.R) AND VARIABLE RESOLUTION (V.R) SETTING. BOLD RESULTS INDICATE THE BEST OUTCOME AMONG THE COMPARISONS. ALL THE RESIDUAL SCALE FACTOR IS 1.0.

| Methods | LGDR | DCTDWTSVD | SSLWM | CIN | HiDDeN | | MuST | | TrustMark | | VideoSeal | | VaRiA | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | - | - | - | - | I | II | I | II | Q | P | 0.0 | 1.0 | I | II |
| Test Setting | F.R | F.R | F.R | F.R | F.R | V.R | F.R | V.R | V.R | V.R | V.R | V.R | F.R | V.R |
| PSNR | 40.78 | 36.52 | 36.89 | 42.56 | 36.89 | 34.56 | 43.24 | 39.56 | 43.26 | 45.69 | **49.65** | 46.13 | 45.98 | 44.25 |
| SSIM | 0.9854 | 0.9598 | 0.9493 | 0.9792 | 0.9493 | 0.9191 | 0.9883 | 0.9603 | 0.9900 | 0.9938 | **0.9991** | 0.9974 | 0.9957 | 0.9892 |
| LPIPS | 0.010 | 0.016 | 0.012 | 0.008 | 0.014 | 0.016 | 0.009 | 0.013 | 0.008 | 0.005 | **0.002** | 0.004 | 0.008 | 0.010 |

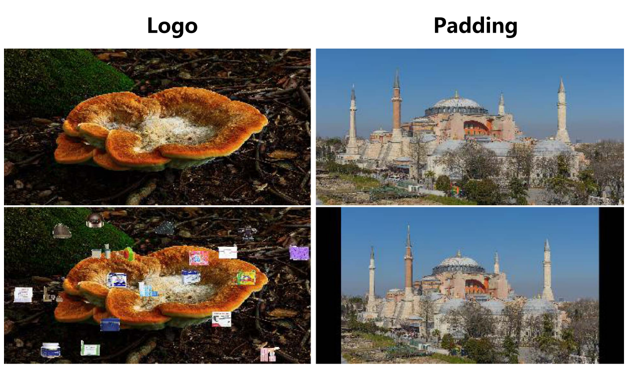



Fig. 4. Visualization of Logo and Padding distortions.

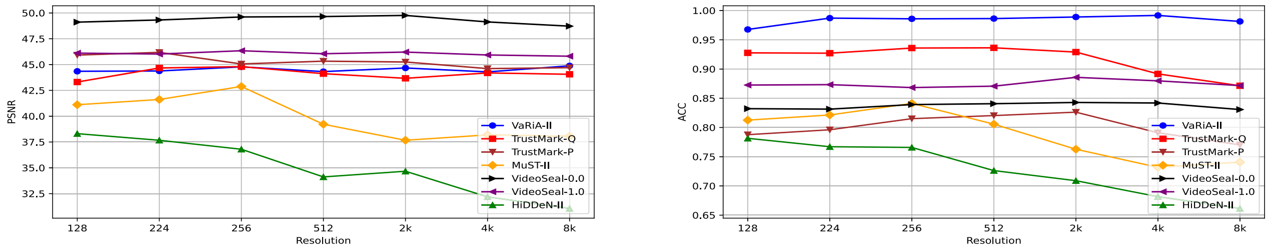
images, compared with rotation and affine transformation. The visualizations of these two perturbations are shown in Fig. 4.

In simple terms, the Logo perturbation simulates adding new icons to a computer desktop, which is common for variable-resolution images used as desktop wallpapers. The Padding perturbation adjusts the image to reach a specific resolution without stretching (i.e., by adding borders or padding).

As shown in Table V, after the fine-tuning in stage II, the same methods demonstrate a significant increase in robustness across various resolution scenarios. In Table V, it is evident that fine-tuning not only enhances the algorithm's robustness but also keeps the visual quality loss within acceptable limits. Notably, our adversarial perturbations did not include the Logo and Padding scenarios, yet VaRiA still exhibits robustness against these perturbations. This phenomenon suggests that VaRiA, implemented with a Transformer architecture, leverages the self-attention mechanism, allowing the model to better focus on the watermark distribution. Unlike convolutional networks, which can only attend to local information, the attention mechanism facilitates interaction between different image blocks. This implicitly introduces a correction mechanism for the watermark, so even with a significant presence of non-watermarked areas in the input image, the impact on performance is not as pronounced as in other methods.

These experimental observations sufficiently demonstrate the versatility of our proposed two-stage training and fine-tuning approach. They also confirm that the VaRiA architecture not only excels in standard scenarios but also achieves excellent robustness and minimal visual loss in complex, variable-resolution environments.

To investigate whether our method can effectively adapt to variable common resolution inputs, we conduct experiments across a comprehensive resolution range from the lowest $128 \times 128$ resolution up to 8 K resolution. For each resolution setting, we employ the same combined noise profile detailed in Table V, where one noise type is randomly selected during each testing iteration. The final performance evaluation is obtained through average accuracy computation across all iterations. The results are shown in Fig. 5. Notably, compared to TrustMark (trained on 2 K images), our method maintains robust performance when processing 8 K images, while TrustMark demonstrates significant performance degradation at 4 K resolution. Furthermore, VideoSeal demonstrated consistent performance across the entire test range. Although it achieved state-of-the-art visual quality ( 4 dB better than VaRiA), it was less robust than VaRiA ( 8%




(a) (b)

Fig. 5. Results of different resolution inputs. (a) Average PSNR testing with different resolution inputs. (b) Average accuracy testing with different resolution inputs.

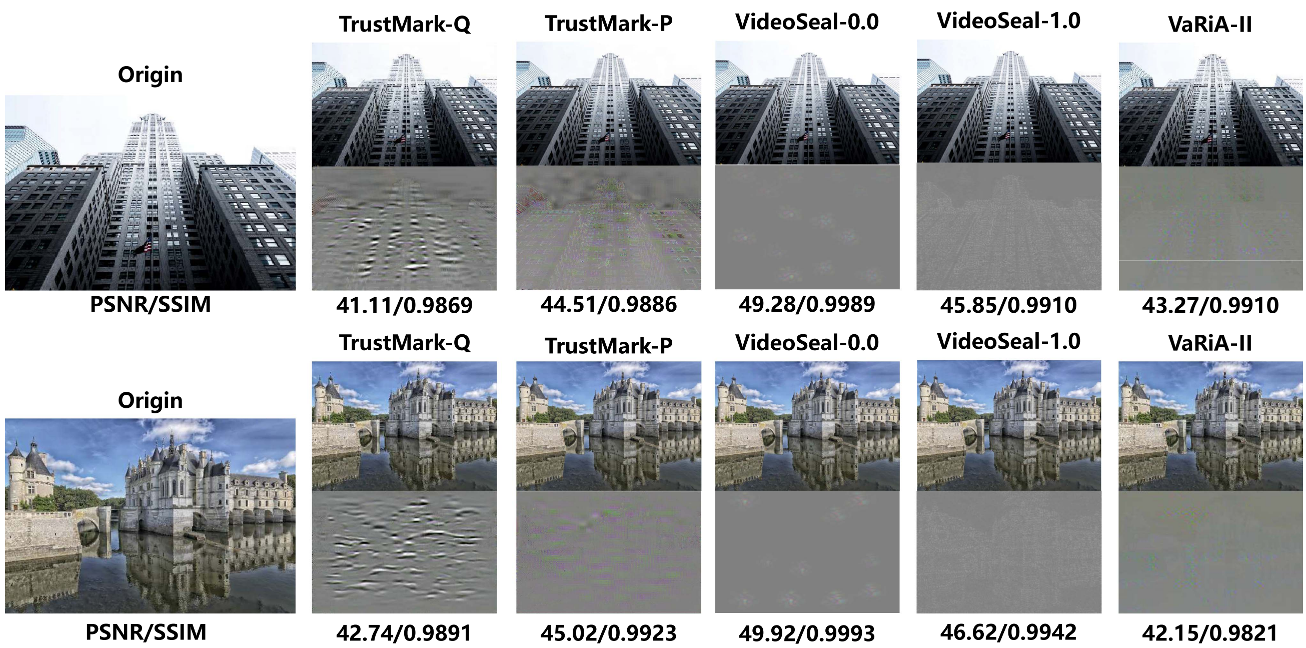



Fig. 6. Watermark visualization of Trustmark, VideoSeal, and VaRiA-II. The first row is the watermarked images, and the second row is the watermark. To enhance the visibility of the watermarking results, we multiplied each watermark by a coefficient $c = 10$.

bit accuracy). This demonstrates that our approach effectively accommodates all currently prevalent resolution configurations.

*3) Watermark Visualization:* As shown in Fig. 7, the watermark distribution generated by VaRiA trained only in stage I notably differs from other methods, exhibiting a strong block-like expression. We believe this is due to the transformation of the image into embeddings, a characteristic of the embedding process itself. Additionally, the powerful information extraction capability of the Transformer architecture enables accurate watermark extraction in scenarios like those shown in Fig. 3, 4. This advantage is demonstrated in the robustness experiments.

As shown in Fig. 6, VaRiA trained through stage II demonstrates a watermark distribution that adheres more closely to the image than TrustMark-Q. As expected, while TrustMark-P and VideoSeals achieve higher PSNR, their extraction accuracy is lower than that of VaRiA, whereas VaRiA successfully unifies TrustMark-P's high fidelity with TrustMark-Q's robustness. Compared to the watermark from Stage I, Stage II results show a more scaled appearance, indicating an adjustment in watermark embedding that aligns better with the image content.

Notably, in models trained after each of the two stages, distinct prominent blocks appear within the watermark. We attribute this to the Attention mechanism, which enriches the watermark and image information, resulting in one or more blocks that effectively represent overall information. This phenomenon aligns with Meta's observations in DINOv2 [41], and optimizing these blocks will be an important focus for our future work.

*4) Real-World Scenario:*

*a) Wallpaper:* A classical application of high-resolution images (e.g., 2 K or 4 K) is as wallpapers or background images. We evaluate the effectiveness of VaRiA in real-world wallpaper usage scenarios. As shown in the Figs. 8 and 9, we embed a 64-bit message into each original image, resulting in two watermarked images with PSNR values of 44.75 dB and 43.79 dB, and SSIM scores of 0.9932 and 0.9908, respectively. We then capture full-screen desktop screenshots and extract

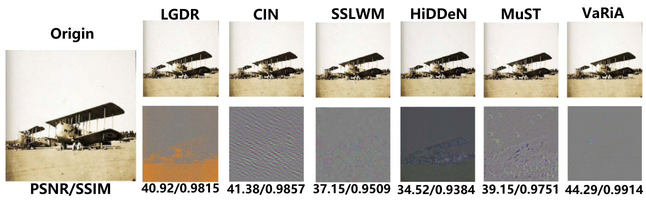



Fig. 7. Watermark visualization of each method evaluated in Table III. The original image is sampled from the MS COCO validation dataset. No external distortions were applied here in order to explicitly illustrate the distinct, native watermark patterns generated by each method. To enhance visibility, we multiplied each extracted watermark by a coefficient $c = 10$.

TABLE VII
THE AVERAGE PSNR (DB) AND SSIM OF EACH METHOD TESTED ON THE VALIDATION SET OF MS COCO, DIV2K, AND FLICKR2K

| $L_{vgg}$ | $L_{LPIPS}$ | $L_{adversial}$ | PSNR | SSIM |
|---|---|---|---|---|
| × | × | × | 33.92 | 0.9158 |
| ✓ | × | × | 36.06 | 0.9384 |
| × | ✓ | × | 35.84 | 0.9306 |
| × | × | ✓ | 37.59 | 0.9521 |
| ✓ | ✓ | × | 36.91 | 0.9508 |
| ✓ | × | ✓ | 39.58 | 0.9677 |
| × | ✓ | ✓ | 42.61 | 0.9719 |
| ✓ | ✓ | ✓ | 44.25 | 0.9892 |

TABLE VIII
THE AVERAGE PSNR (DB) AND SSIM OF EACH METHOD TESTED ON THE VALIDATION SET OF MS COCO, DIV2K, AND FLICKR2K. W/O REPRESENTS THE MODEL DOES NOT UTILIZE MAB.

| Methods | w/o-I | w/o-II | w-I | w-II |
|---|---|---|---|---|
| PSNR | 38.52 | 38.94 | 45.98 | 44.25 |
| SSIM | 0.9754 | 0.9812 | 0.9957 | 0.9892 |

TABLE IX
THE AVERAGE $\overline{ACC}(\%)$ TESTED ON THE VALIDATION SET OF MS COCO, DIV2K AND FLICKR2K. *COMBINED* MEANS RANDOMLY CHOOSING THE CLASSIC DISTORTION TO PROCESS THE WATERMARKED IMAGE.

| Type \ Methods | w/o-I | w/o-II | w-I | w-II |
|---|---|---|---|---|
| Combined | 95.32 | 96.59 | 98.57 | 98.82 |
| Logo | 70.25 | 97.91 | 74.23 | 99.73 |
| Padding | 64.89 | 95.21 | 65.61 | 97.91 |

TABLE X
THE AVERAGE COMPUTATION COST TEST UNDER VARIABLE RESOLUTION SETTING IN THE EMBEDDING AND EXTRACTION PROCESS

| Methods | VaRiA-w/o | VaRiA | TrustMark-Q |
|---|---|---|---|
| # Params (M) | 36.01 | 37.18 | 30.86 |
| GFLOPs | 21.19 | 23.07 | 28.80 |
| Embedding Time(s) | 0.0029 | 0.0034 | 0.0023 |
| Extraction Time(s) | 0.0052 | 0.0061 | 0.0045 |
| PSNR(dB) | 38.94 | 44.25 | 43.26 |
| $\overline{ACC}(\%)$ under Combined | 96.59 | 98.82 | 92.61 |

the watermarks, achieving accuracies of 98.44% and 100%. These results clearly demonstrate the effectiveness of VaRiA in practical wallpaper deployment scenarios.

*b) Social Media:* As shown in Fig. 9, we published a watermarked image on a social media platform,[3] cropped it to remove the platform's visible watermark and then extracted our embedded watermark from the cropped image to verify robustness in another variable-resolution application scenario. The original image contains 64 bits watermark. In this sample, the PSNR/SSIM of watermarked images reached 44.52/0.9913 and 46.19/0.9984, and the watermark extraction accuracy from the screenshot was 98.44% and 98.44%. It is also worth noting that some platforms may automatically add an alpha channel or apply image compression to uploaded images. For the former, since VaRiA accepts only three-channel inputs, we manually removed the alpha channel before processing. For the latter, VaRiA's inherent robustness to compression allows it to handle such scenarios without additional adjustments.

The above experimental results demonstrate that VaRiA is capable of effectively handling common real-world application scenarios.

### C. Ablation Study

*1) Message Attention Block:* To validate the effectiveness of MAB, we tested a model that does not utilize MAB, replacing it with classic Attention based on the experimental setup described earlier.

Comparing the results in Fig. 10(a) with those in Fig. 7, it is evident that the watermark is more prominent and the visual quality is worse without using MAB. In Fig. 10(b), a comparison with Fig. 6 reveals that the watermark, in the absence of MAB, is less capable of representing the image's detailed information, with blocky artifacts becoming more pronounced. This is also reflected in the PSNR and SSIM results in Table VIII. These findings indicate that the introduction of MAB enables the model to better focus on image details and the relationships between different image blocks.

As shown in Table IX, we conducted a simple test regarding the methods' robustness under combinations of perturbations such as Crop, Gaussian Blur, and the previously mentioned Logo and Padding (listed as the Combined item). In most cases, the methods' robustness does not significantly decline compared to the previously presented results. This indicates that models utilizing the classic Attention mechanism can still represent sufficiently robust watermark features to meet the needs of most scenarios. However, when dealing with distortions encountered in real-world scenarios, MAB demonstrates superior capability in representing and extracting watermark signals compared to traditional attention mechanisms. We recorded the computation cost during the embedding and extraction process compared with TrustMark, shown in Table X. While MAB adds 0.0027 seconds to embedding and extraction, it improves PSNR by 1 dB and robustness by 6%. This slight overhead is a worthwhile trade-off for the performance gains.

*2) Loss Function:* To validate the effectiveness of our vision-related constraint loss functions, we evaluate visual quality while maintaining the extraction accuracy at 98% under the

[3]https://x.com

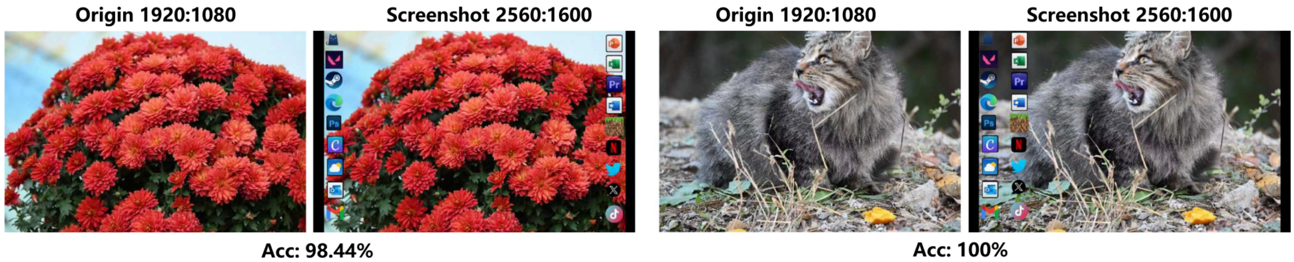
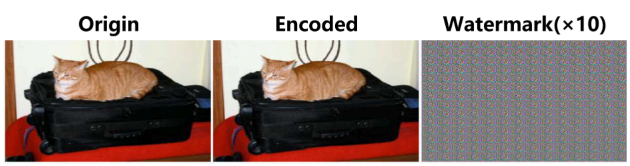
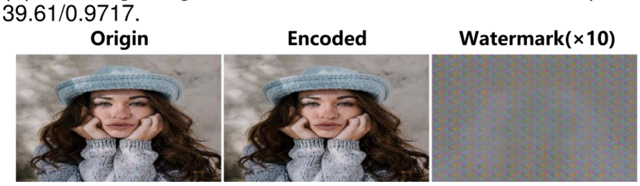



Fig. 8. The example of the watermarked image used for wallpaper. The original image is captured with a Nikon Z30 with a resolution of 1920 × 1080. The screenshot of the background on a display with a resolution of 2560 × 1600.


Fig. 9. The example of the watermarked image used for social media. The original image is captured with a Nikon Z30 with a resolution of 853 × 1280.




(a) Training stage I. The PSNR/SSIM of the example is 39.61/0.9717.




(b) Training stage II. The PSNR/SSIM of the example is 38.92/0.9741.

Fig. 10. Visualization of VaRiA-w/o after different stage training. **w/o** represents the model does not utilize MAB.

combined noise conditions established in previous experiments. As shown in Table VII, only when incorporating all designed loss components does the watermarking algorithm achieve both robustness and superior imperceptibility simultaneously. Moreover, we observed significant improvements in watermark imperceptibility under both scenarios I and II. This demonstrates two key contributions: (1) The adversarial learning framework with a discriminative network effectively optimizes watermark patterns to enhance visual transparency, and (2) the incorporation of LPIPS-based feature-level perceptual constraints enables the model to generate watermark patterns that better align with image content characteristics.

TABLE XI
THE PERFORMANCE WITH DIFFERENT MESSAGE LENGTHS

| Message Length(bits) | PSNR | SSIM | $\overline{ACC}(\%)$ |
|---|---|---|---|
| 32 | 47.85 | 0.9967 | 99.01 |
| 64 | 44.25 | 0.9892 | 98.82 |
| 128 | 41.06 | 0.9714 | 95.71 |
| 196 | 37.52 | 0.9481 | 93.53 |

*3) Message Length:* To explore the boundary of the model's capacity, we trained and tested the model with different watermark lengths. As shown in Table XI, VaRiA is capable of embedding 128-bit watermarks while maintaining a PSNR above 40 dB and keeping the extraction accuracy degradation within 5 percentage points. However, we ultimately chose 64-bit watermarks as our default experimental setting, as this configuration offers a desirable balance among visual quality, robustness, and capacity. For example, in practical applications, a raw watermark is typically encoded into a 64-bit message using BCH coding before embedding. Our experimental results show that VaRiA produces at most 3-bit errors in most cases, which allows for up to 45 bits of effective payload when BCH coding is applied, sufficient for real-world usage.

*4) Number of Patches:* To investigate the appropriate number of patches used for constraining visual quality in Stage II, we evaluate the model's visual fidelity and per-epoch training time while maintaining comparable robustness. As shown in Table XII, increasing the number of patches results in improved

TABLE XII
THE PERFORMANCE WITH DIFFERENT NUMBERS OF PATCHES. TIME: AVERAGE TRAINING TIME PER STEP

| Number of Patches | PSNR | SSIM | Time(ms) |
|---|---|---|---|
| 1 | 42.41 | 0.9882 | 1.7 |
| 2 | 42.97 | 0.9905 | 2.6 |
| 4 | 44.25 | 0.9892 | 5.0 |
| 8 | 46.51 | 0.9963 | 9.7 |
| 12 | 47.30 | 0.9984 | 15.2 |

visual quality. This is because the model receives more undistorted visual features, enabling it to better preserve image quality during watermark embedding. However, the training time increases almost quadratically with the number of patches. To balance training efficiency and effectiveness, we ultimately set the number of patches to $n = 4$.

## V. LIMITATION

As detailed in Table XI, introducing the MAB slightly increases the parameter size to 37.18 M. Interestingly, VaRiA operates with significantly lower computational complexity (23.07 GFLOPs) compared to the CNN-based TrustMark (28.80 GFLOPs). The observed 17% increase in inference time is not due to mathematical complexity, but rather the memory-bandwidth-bound operations typical of Transformer architectures (e.g., LayerNorm and feature concatenation). Given the substantial performance gains (+1 dB PSNR and +6% robustness), this minor latency overhead is a highly justified trade-off. Although there was a significant performance improvement, optimizing the model to maintain performance with fewer parameters is another challenge that needs to be addressed. In addition, there is a growing body of research focused on watermark removal and forgery. We conducted watermark removal experiments on VaRiA using the VAE component of Stable Diffusion v1.4 [42]. The results show that even when the visual quality is degraded to an average PSNR of 39.81 and SSIM of 0.9762, the average watermark extraction accuracy remains at 96%, indicating that VaRiA is resilient to certain simple watermark removal attacks. We plan to further investigate its robustness against more sophisticated removal and forgery methods in future.

## VI. CONCLUSION

In this paper, we highlighted the significant visual quality loss and insufficient robustness that current digital watermarking technologies face in variable resolution scenarios. To address these issues, we proposed a novel Transformer-based watermarking framework, VaRiA, which incorporates Message Attention Blocks to capture the relationship between pixels and reconstruct watermarked images, thereby optimizing the quality of the generated watermarked images. Additionally, we introduced a dual-stage training strategy to accommodate images of variable resolutions better. VaRiA demonstrates strong performance across various resolutions and perturbations.

## REFERENCES

[1] J. Zhu, R. Kaplan, J. Johnson, and L. Fei-Fei, "Hidden: Hiding data with deep networks," in *Proc. 15th Eur. Conf. Comput. Vis.*, 2018, pp. 682–697.
[2] M. Tancik, B. Mildenhall, and R. Ng, "StegaStamp: Invisible hyperlinks in physical photographs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 2117–2126.
[3] G. Wang et al., "Must: Robust image watermarking for multi-source tracing," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 6, pp. 5364–5371.
[4] R. Ma et al., "Towards blind watermarking: Combining invertible and non-invertible mechanisms," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 1532–1542.
[5] P. Fernandez, A. Sablayrolles, T. Furon, H. Jégou, and M. Douze, "Watermarking images in self-supervised latent spaces," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2022, pp. 3054–3058.
[6] D. Podell et al., "SDXL: Improving latent diffusion models for high-resolution image synthesis," in *Proc. Int. Conf. Learn. Representations*, 2023, pp. 1862–1874.
[7] D. Liu et al., "Lumina-MGPT: Illuminate flexible photorealistic text-to-image generation with multimodal generative pretraining," 2024, *arXiv:2408.02657*.
[8] A. Brock, J. Donahue, and K. Simonyan, "Large scale GAN training for high fidelity natural image synthesis," 2018, *arXiv:1809.11096*.
[9] Z. Jia, H. Fang, and W. Zhang, "MBRS: Enhancing robustness of DNN-based watermarking by mini-batch of real and simulated JPEG compression," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 41–49.
[10] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.
[11] S. Lu, Z. Zhou, J. Lu, Y. Zhu, and A. W.-K. Kong, "Robust watermarking using generative priors against image editing: From benchmarking to advances," in *Proc. Int. Conf. Learn. Representations*, vol. 2025, 2025, pp. 83902–83936.
[12] A. Dosovitskiy et al., "An image is worth $16 \times 16$ words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[13] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7132–7141.
[14] H. Fang, Z. Jia, Z. Ma, E.-C. Chang, and W. Zhang, "Pimog: An effective screen-shooting noise-layer simulation for deep-learning-based watermarking network," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 2267–2275.
[15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
[16] T. Sander, P. Fernandez, A. Durmus, T. Furon, and M. Douze, "Watermark anything with localized messages," in *Proc. Int. Conf. Learn. Representations*, vol. 2025, 2025, pp. 79569–79599.
[17] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4015–4026.
[18] R. Hu et al., "Mask image watermarking," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 38, 2026, pp. 146313–146346.
[19] X. Kang, J. Huang, Y. Q. Shi, and Y. Lin, "A DWT-DFT composite watermarking scheme robust to both affine transform and JPEG compression," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 13, no. 8, pp. 776–786, Aug. 2003.
[20] Z. Ma, W. Zhang, H. Fang, X. Dong, L. Geng, and N. Yu, "Local geometric distortions resilient watermarking scheme based on symmetry," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 12, pp. 4826–4839, Dec. 2021.
[21] H. Fang, W. Zhang, H. Zhou, H. Cui, and N. Yu, "Screen-shooting resilient watermarking," *IEEE Trans. Inf. Forensics Secur.*, vol. 14, no. 6, pp. 1403–1418, Jun. 2019.
[22] J. Jia, Z. Gao, D. Zhu, X. Min, G. Zhai, and X. Yang, "Learning invisible markers for hidden codes in offline-to-online photography," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 2273–2282.
[23] Y. Luo, T. Zhou, S. Cui, Y. Ye, F. Liu, and Z. Cai, "Fixing the double agent vulnerability of deep watermarking: A patch-level solution against artwork plagiarism," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 3, pp. 1670–1683, Mar. 2024.
[24] P. Fernandez, H. Elsahar, I. Z. Yalniz, and A. Mourachko, "Video seal: Open and efficient video watermarking," 2024,*arXiv:2412.09492*.
[25] T. Bui, S. Agarwal, and J. Collomosse, "Trustmark: Universal watermarking for arbitrary resolution images," 2023,*arXiv:2311.1829*.
[26] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[27] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014,*arXiv:1409.1556*.
[28] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 586–595.
[29] U. Demir and G. B. Ünal, "Patch-based image inpainting with generative adversarial networks," 2018,*arXiv:1803.07422*.
[30] T.-Y. Lin et al., "Microsoft CoCo: Common objects in context," in *Proc. 13th Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
[31] E. Agustsson and R. Timofte, "Ntire 2017 challenge on single image super-resolution: Dataset and study," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2017, pp. 126–135.
[32] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2017, pp. 136–144.
[33] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8026–8037.
[34] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017,*arXiv:1711.05101*.
[35] K. Navas, M. C. Ajay, M. Lekshmi, T. S. Archana, and M. Sasikumar, "DWT-DCT-SVD based watermarking," in *Proc. 3rd Int. Conf. Commun. Syst. Softw. Middleware Workshops*, 2008, pp. 271–274.
[36] I. J. Cox, J. Kilian, T. Leighton, and T. Shamoon, "Secure spread spectrum watermarking for images, audio and video," in *Proc. 3rd IEEE Int. Conf. Image Process.*, 1996, vol. 3, pp. 243–246.
[37] I. J. Cox, J. Kilian, F. T. Leighton, and T. Shamoon, "Secure spread spectrum watermarking for multimedia," *IEEE Trans. Image Process.*, vol. 6, no. 12, pp. 1673–1687, Dec. 1997.
[38] L. Pérez-Freire and F. Pérez-González, "Spread-spectrum watermarking security," *IEEE Trans. Inf. Forensics Secur.*, vol. 4, no. 1, pp. 2–24, Mar. 2009.
[39] A. Almohammad and G. Ghinea, "Stego image quality and the reliability of PSNR," in *Proc. 2nd Int. Conf. Image Process. Theory, Tools Appl.*, 2010, pp. 215–220.
[40] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[41] T. Darcet, M. Oquab, J. Mairal, and P. Bojanowski, "Vision transformers need registers," in *Proc. Int. Conf. Learn. Representations*, 2024, pp. 2632–2652.
[42] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

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